

GARMENTFLOW

Garments Supply Chain Intelligence & Decision Support System

PROJECT PROPOSAL / PROJECT REPORT

DEVELOPER	Umma Salma Mosammad Samima Aktar
Trainee ID	1295920
Batch	WDPF/CCSL-M/71/01
Program	IsDB-BISEW IT Scholarship Project
PROJECT SUPERVISOR	Tawhid Imdad
Cell No.	01985426948
PROJECT CONSULTANT	Md. Moshaidul Islam
Organization	WDPF, IsDB-BISEW IT Scholarship Project

IsDB-BISEW IT Scholarship Project
Dhaka, Bangladesh

. Project Proposal To

Project Consultant

WDPF, IsDB-BISEW IT Scholarship Project

IDB Bhaban, Sher-e-Bangla Nagar, Dhaka, Bangladesh

Subject: Submission of Project Proposal for GarmentFlow – Garments Supply Chain Intelligence & Decision Support System

Dear Sir,

I am pleased to submit my project proposal entitled "GarmentFlow – Garments Supply Chain Intelligence & Decision Support System", prepared as part of the WDPF, IsDB-BISEW IT Scholarship Project.

This proposal presents the design and development of a web-based supply chain management and decision support system for garments manufacturing businesses. The proposed system integrates buyer order management, demand forecasting, supply chain planning, material requirements, inventory management, procurement, warehouse operations, production, sales, invoicing, payments, delivery, and business reporting into a centralized platform.

The system provides five role-based dashboards for the CEO, Supply Chain Manager, Production Manager, Procurement Manager, and Warehouse Manager. These dashboards provide relevant KPIs, reports, alerts, forecasts, and operational information to support effective business decisions.

The system aims to improve supply chain visibility, reduce material shortages, improve inventory management, monitor production and procurement activities, and support data-driven decision-making.

Thank you for your time and consideration.

Sincerely,

Umma Salma Mosammad Samima Aktar

Trainee ID: 1295920

Batch: WDPF/CCSL-M/71/01

1. Introduction

GarmentFlow – Garments Supply Chain Intelligence & Decision Support System is a web-based enterprise application designed to manage and monitor the major operations of a garments manufacturing business. A garments manufacturing business involves several interconnected activities. A buyer places an order, the company analyzes the demand, plans the required materials, checks inventory, purchases materials when necessary, stores them in the warehouse, produces garments, stores finished goods, manages sales, generates invoices, receives payments, and finally delivers the products. GarmentFlow connects these activities through a centralized system.

2. Background

The garments industry has a complex supply chain involving buyers, suppliers, raw materials, warehouses, production units, finished goods, sales, and deliveries. In many organizations, different departments use separate spreadsheets or manual processes to maintain information. As a result, management may face difficulties in obtaining accurate and timely information about orders, inventory, procurement, production, and sales. GarmentFlow is proposed to solve these problems by connecting the major business processes into one centralized digital platform.

3. Problem Statement

Garments manufacturing businesses face several operational and supply chain challenges: difficulty tracking buyer orders; manual demand and material planning; lack of centralized supply chain information; material shortages during production; excess or insufficient inventory; difficulty monitoring warehouse stock; delayed procurement activities; difficulty monitoring supplier performance; production delays; lack of real-time production information; manual invoice and payment management; difficulty tracking deliveries; lack of centralized business reports; lack of real-time management dashboards; difficulty identifying supply chain risks; and delayed decision-making due to lack of accurate information.

4. Objectives

The main objectives are to develop a centralized garments supply chain management system; manage buyer information and buyer orders; support demand forecasting; perform supply chain planning; calculate material requirements; monitor raw material and finished goods inventory; identify material shortages; manage suppliers and procurement activities; monitor warehouse operations; manage production planning and progress; manage sales transactions; generate invoices automatically; track payments; manage delivery operations; generate business reports; provide role-based dashboards; generate alerts for important business situations; support data-driven management decisions; and improve overall supply chain efficiency.

5. Scope of the Project

Master Data: User Management • Role Management • Buyer Management • Customer Management • Product Management • Category Management • Size Management • Color Management

Order Management: Buyer Orders • Order Items • Order Status • Order History • Delivery Date

Planning: Demand Forecasting • Supply Planning • Material Requirement Planning

Inventory & Warehouse: Warehouse Management • Raw Material Inventory • Finished Goods Inventory • Stock In • Stock Out • Stock Transfer • Stock Adjustment • Stock Report

Procurement: Supplier Management • Purchase Requisition • Purchase Orders • Purchase Order Items • Goods Receipt • Purchase Report

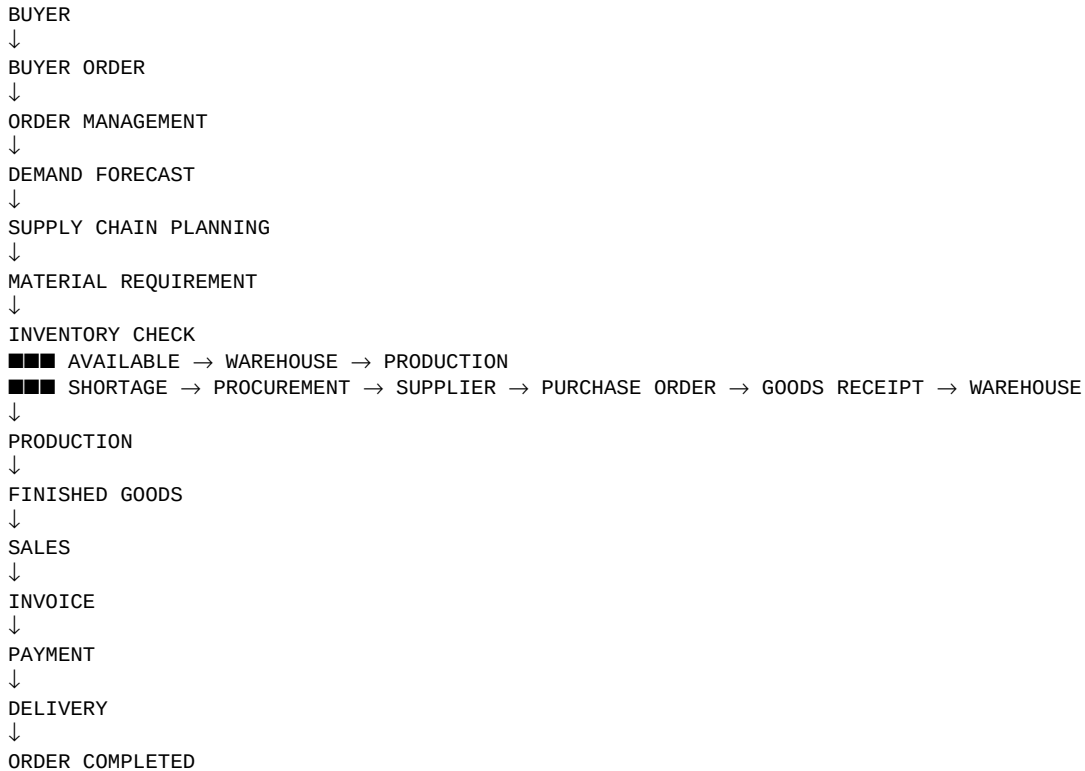
Production: Production Planning • Production Orders • Production Progress • Material Consumption • Finished Goods • Production Reports

Sales & Finance: Sales Orders • Sales Items • Invoice • Invoice Items • Payments • Profit Report

Delivery: Delivery Management • Delivery Status • Delivery Tracking • Delivery Report

Intelligence & Reporting: Dashboards • KPIs • Alerts • Forecasting • Business Reports • Decision Support

6. Business Workflow



7. Business Logic

Step 1 — Buyer Order — Buyer places an order containing buyer, product, quantity, order date, required delivery date, and order status.

Step 2 — Demand Forecast — Buyer orders and historical order/sales data are analyzed to generate demand forecasts.

Step 3 — Supply Planning — Forecasted demand and confirmed orders are used to create the supply plan.

Step 4 — Material Requirement — The system calculates how much raw material is required for planned production.

Step 5 — Inventory Check — The system checks whether required material is available. If available, production can proceed; if there is a shortage, procurement is initiated.

Step 6 — Procurement — Material shortages generate purchase requisitions and purchase orders. Suppliers provide the required materials and goods receipt is recorded.

Step 7 — Warehouse — Received materials are added to warehouse inventory.

Step 8 — Production — Raw materials are issued to production. On completion, finished goods are added to finished-goods inventory.

Step 9 — Sales — Finished goods are sold through sales orders.

Step 10 — Invoice & Payment — A confirmed sales order generates an invoice and payment is recorded as paid, partial, or unpaid.

Step 11 — Delivery — After order/payment confirmation, delivery is created, tracked, and finally marked completed.

8. User Roles & Dashboards

Role	Dashboard	Main Responsibilities
CEO	Executive Dashboard	Overall KPIs, revenue, sales, purchases, profit, inventory value, production, payments, deliveries
Supply Chain Manager	Supply Chain Dashboard	Demand forecast, supply planning, material requirements, inventory availability, shortages, stock
Production Manager	Production Dashboard	Production plan, production orders, progress, material consumption, finished goods
Procurement Manager	Procurement Dashboard	Suppliers, purchase requisitions, purchase orders, goods receipt, supplier performance
Warehouse Manager	Warehouse Dashboard	Warehouses, raw materials, finished goods, stock in/out, transfers, adjustments, stock reports

9. Functional Requirements

Authentication Module: User Registration • Login • Logout • Password Change • Role-Based Access • Permission Management

Buyer Module: Add Buyer • View Buyer • Edit Buyer • Delete Buyer • Buyer Order History

Product Module: Add Product • Edit Product • Delete Product • Product Category • Product Variant • Size • Color

Order Module: Create Buyer Order • Order Items • Quantity • Delivery Date • Order Status • Order History

Demand Forecast Module: Historical Data • Demand Forecast • Forecast Quantity • Forecast Comparison • Forecast Report

Supply Chain Module: Supply Planning • Material Requirement • Inventory Check • Material Shortage Detection • Supply Risk

Inventory Module: Stock In • Stock Out • Stock Transfer • Stock Adjustment • Inventory Tracking • Low Stock Alert • Stock Report

Procurement Module: Supplier Management • Purchase Requisition • Purchase Order • Goods Receipt • Supplier Performance • Purchase Report

Production Module: Production Plan • Production Order • Production Progress • Material Consumption • Finished Goods • Production Report

Sales Module: Customer Management • Sales Order • Sales Items • Quantity • Price • Discount • Sales History

Invoice Module: Automatic Invoice Generation • Invoice Number • Invoice Items • Subtotal • Discount • Grand Total • Print Invoice

Payment Module: Payment Recording • Paid • Partial • Unpaid • Outstanding Payment • Payment History

Delivery Module: Delivery Creation • Delivery Address • Delivery Date • Delivery Status • Delivery Tracking

Report Module: Invoice Report • Stock Report • Sales Report • Purchase Report • Profit Report • Production Report • Procurement Report • Inventory Report • Payment Report • Delivery Report

10. Database Modules

User & Access Management: users • roles • permissions • role_permissions

Buyer & Product Management: buyers • categories • products • product_variants • sizes • colors

Order Management: orders • order_items • order_status_history

Demand Forecasting: demand_forecasts • forecast_items

Supply Chain Planning: supply_plans • material_requirements

Warehouse & Inventory: warehouses • inventory • stock_movements • stock_adjustments

Procurement: suppliers • purchase_requisitions • purchase_orders • purchase_order_items • goods_receipts

Production: production_orders • production_items • production_progress • material_consumptions

Sales: customers • sales_orders • sales_order_items

Invoice & Payment: invoices • invoice_items • payments

Delivery: deliveries • delivery_items

Alerts & Notifications: notifications • alerts

11. Reports & Analytics

Invoice Report — Invoice number, buyer/customer, invoice date, total amount, payment status.

Stock Report — Product/material, warehouse, opening stock, stock in, stock out, closing stock.

Sales Report — Sales date, product, quantity, price, revenue.

Purchase Report — Supplier, material, quantity, purchase cost, purchase date.

Profit Report — Sales Revenue – Cost of Goods / Production Cost = Gross Profit.

Production Report — Planned quantity, produced quantity, pending quantity, production efficiency.

Procurement Report — Purchase requisition, purchase order, ordered quantity, received quantity, pending quantity.

Inventory Report — Raw material, finished goods, inventory value, low stock.

Payment Report — Paid, partial, unpaid, outstanding.

Delivery Report — Pending, shipped, delivered, delayed.

12. Technology Used

Frontend: React.js • Vite • HTML5 • CSS3 • Bootstrap 5 / Tailwind CSS • JavaScript • Axios • Recharts

Backend: PHP • Laravel

Database: MySQL

Authentication: Laravel Sanctum

Development Tools: Visual Studio Code • XAMPP • Git • GitHub • Postman

13. Proposed Project Structure

```
GarmentFlow/
├── backend/ ← Laravel API
│   ├── app/
│   │   ├── Http/Controllers/
│   │   ├── Models/
│   │   ├── Services/
│   │   ├── database/
│   │   ├── migrations/
│   │   ├── seeders/
│   │   ├── factories/
│   │   └── routes/api.php
│   ├── .env
│   └── frontend/ ← React
│       ├── src/
│       ├── components/
│       ├── layouts/
│       ├── pages/
│       ├── auth/
│       ├── dashboard/
│       ├── buyers/
│       ├── products/
│       ├── orders/
│       ├── forecasts/
│       ├── supply-chain/
│       ├── inventory/
│       ├── procurement/
│       ├── production/
│       ├── sales/
│       ├── invoices/
│       ├── payments/
│       ├── deliveries/
│       ├── reports/
│       ├── services/
│       ├── context/
│       ├── hooks/
│       ├── routes/
│       └── database/
```

14. Development Plan & Estimated Timeline

Phase	Activity	Duration
1	Requirement Analysis & Project Architecture	3 Days
2	Database Design & Implementation	5 Days
3	Authentication & Role Management	3 Days
4	React Layout & 5 Dashboards	5 Days
5	Buyer, Product & Order Management	4 Days
6	Demand Forecast & Supply Planning	5 Days
7	Inventory & Warehouse Management	5 Days
8	Supplier & Procurement Management	4 Days
9	Production Management	4 Days
10	Sales, Invoice & Payment	4 Days
11	Delivery Management	3 Days
12	Reports & Analytics	4 Days
13	Alerts & Decision Support	3 Days
14	Integration & Testing	4 Days
15	Documentation & Deployment	3 Days
Total		59 Days

15. Expected Outcomes

- Centralized garments supply chain management
- Better buyer order management
- Improved demand forecasting
- Accurate material requirement planning
- Reduced material shortages
- Better inventory visibility
- Efficient procurement management
- Improved supplier monitoring
- Better production tracking
- Effective warehouse management
- Automated invoice generation
- Better payment tracking
- Better delivery management

- Real-time management dashboards
- Automated business reports
- Supply chain risk alerts
- Better management decision-making

16. Future Enhancements

- AI-based Demand Forecasting
- Machine Learning-based Demand Prediction
- AI-based Supply Chain Risk Prediction
- Automated Purchase Recommendations
- Advanced Supplier Performance Analysis
- Barcode Integration
- QR Code Integration
- RFID-based Inventory Management
- Mobile Application
- SMS & Email Notifications
- Live Delivery Tracking
- Multi-Warehouse Support
- Multi-Company Support
- Advanced Business Intelligence
- Accounting Software Integration
- AI-based Production Optimization

17. Conclusion

GarmentFlow – Garments Supply Chain Intelligence & Decision Support System is designed to provide a centralized digital solution for managing the supply chain and operational activities of garments manufacturing businesses.

The system connects the complete business process starting from Buyer Order → Demand Forecast → Supply Planning → Material Requirement → Inventory Check → Procurement → Warehouse → Production → Finished Goods → Sales → Invoice → Payment → Delivery.

The five role-based dashboards provide relevant information to the CEO, Supply Chain Manager, Production Manager, Procurement Manager, and Warehouse Manager according to their responsibilities.

The system also includes Invoice, Stock Report, Sales Report, Purchase Report, Profit Report, Production Report, Procurement Report, Inventory Report, Payment Report, and Delivery Report to provide comprehensive business visibility.

The main objective of GarmentFlow is not simply to digitize existing records. It is designed to analyze operational information and provide KPIs, forecasts, alerts, reports, and decision-support information so that management can make faster and more informed decisions.

Overall, GarmentFlow will provide a practical and scalable solution for improving garments supply chain visibility, inventory control, procurement, production management, reporting, and business decision-making.